Solutions to Review Questions for Lectures 5 and 6

2025-2-6

1. Knowing y helps to predict x in Plot A, indicating an association. Plot B does not show an association between x and y. Plots C and D show an association where x helps to predict y.

2. Based on the definition given in class, $corr(x, y) = \frac{cov(x,y)}{sd(x)sd(y)}$, where cov refers to the covariance. Rewriting the covariance gives:

$$\begin{aligned} cov(x, y) &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n} \\ &= \frac{\sum_{i=1}^{n} x_i y_i - x_i\bar{y} - \bar{x}y_i + \bar{x}\bar{y}}{n} \\ &= \frac{\sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i - \bar{x}\sum_{i-1}^{n} y_i + n\bar{x}\bar{y}}{n} \\ &= \frac{\sum_{i=1}^{n} x_i y_i - \bar{y}n\bar{x} - \bar{x}n\bar{y} + n\bar{x}\bar{y}}{n} \\ &= \overline{xy} - \bar{x}\bar{y} \end{aligned}$$

We can simplify the denominator using the short cut formula as:

$$\begin{aligned} sd(x)sd(y) &= \sqrt{\overline{x^2} - \bar{x}^2}\sqrt{\overline{y^2} - \bar{y}^2} \\ &= \sqrt{(\overline{x^2} - \bar{x}^2)(\overline{y^2} - \bar{y}^2)} \end{aligned}$$

This proves the formula.

3. Changes in scale (subtracting 32 and multiplying by 5/9) do not change correlation, and so the correlation would be 0.97.

4. It would still be 0.97 because changes in scale do not change correlation.

5. A rough estimate for the SD for both Detroit and Chicago is $(85 - (-5))/4 = 22.5$.

6. We can calculate the covariance as $cov(Detroit, Chicago) = r * SD(Detroit) * SD(Chicago) = 0.97 * 22.5 * 22.5 = 491$.

7. Adding 32 does not change the standard deviation but multiplying by 5/9 multiplies the standard deviation by 5/9. Hence the covariance will be $0.97 * (22.5 * 5/9) * (22.5 * 5/9) = 152$.

8. 0.9 : upper left, 0.75 : lower right, 0.5 : upper right, 0 : lower left.

9. Correlation is not a good measure of the strength of a relationship when the relationship is not linear. Some examples of non linear relationships would be between the area and perimeter of a square, between age and blood pressure, and between pressure and volume (Boyle's Law states that pressure is proportional to the reciprocal of volume).

10. Correlation tends to increase when you combine datasets with similar relationships but where the ranges are different in the two datasets. Height and weight will tend to be bigger at age 12 at age 8, but the relationship is similar. You would expect that the correlation would increase when combining the datasets, and so $c > a, c > b$, and hence \$c>(a+b)/2